

UNIVERSIDAD DE GUADALAJARA

Centro Universitario de Ciencias Exactas e Ingenierías



ALU inst

Por:

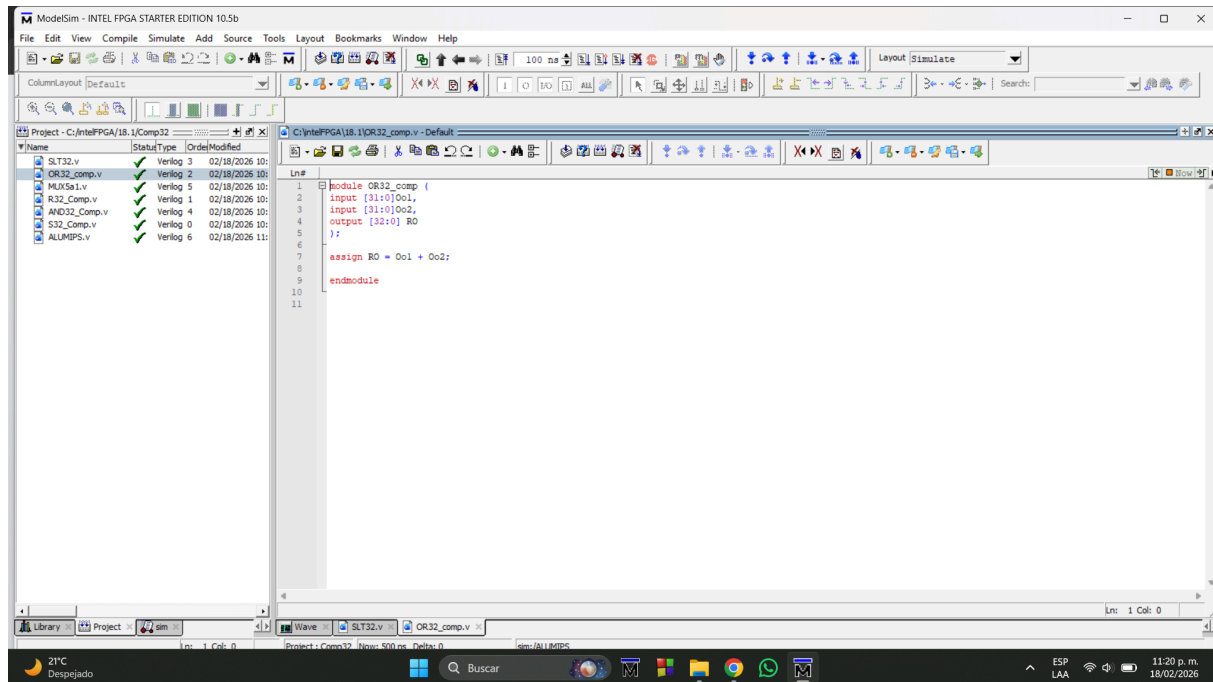
ESTEFANIA NAVARRO MENDOZA

Docente:

JORGE ERNESTO LOPEZ ARCE DELGADO

1. Códigos

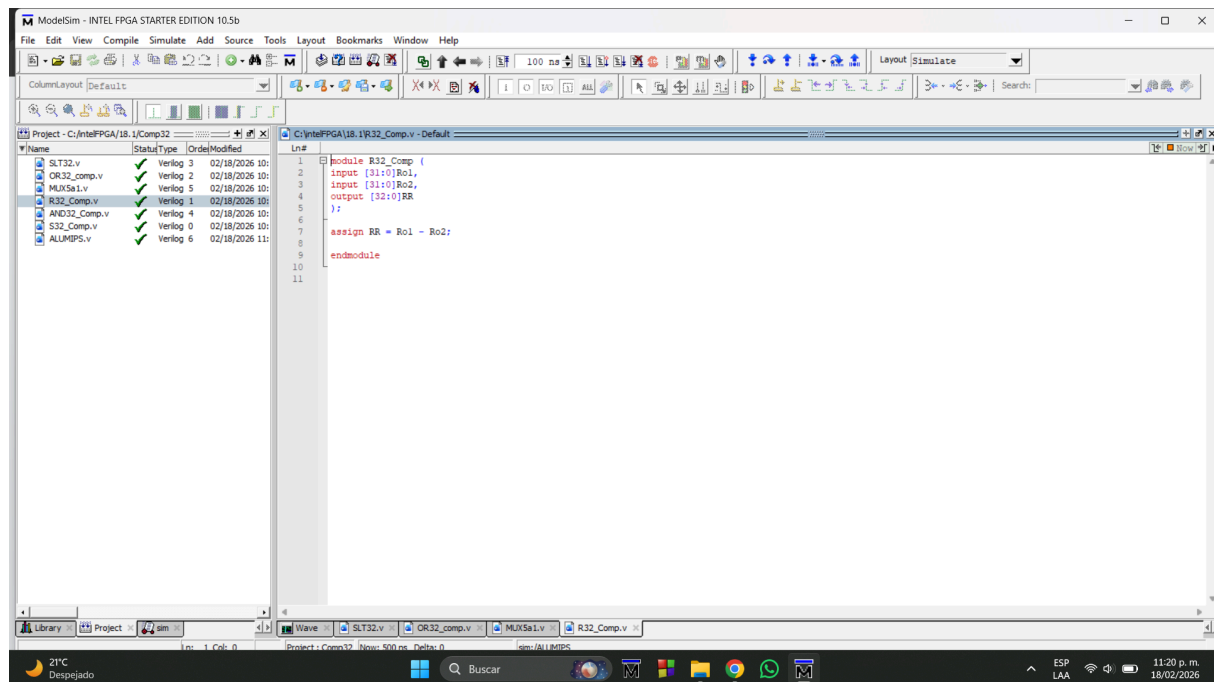
OR



```
module OR32_comp (  
input [31:0]Oo1,  
input [31:0]Oo2,  
output [32:0] RO  
);
```

```
assign RO = Oo1 + Oo2;  
endmodule
```

Resta

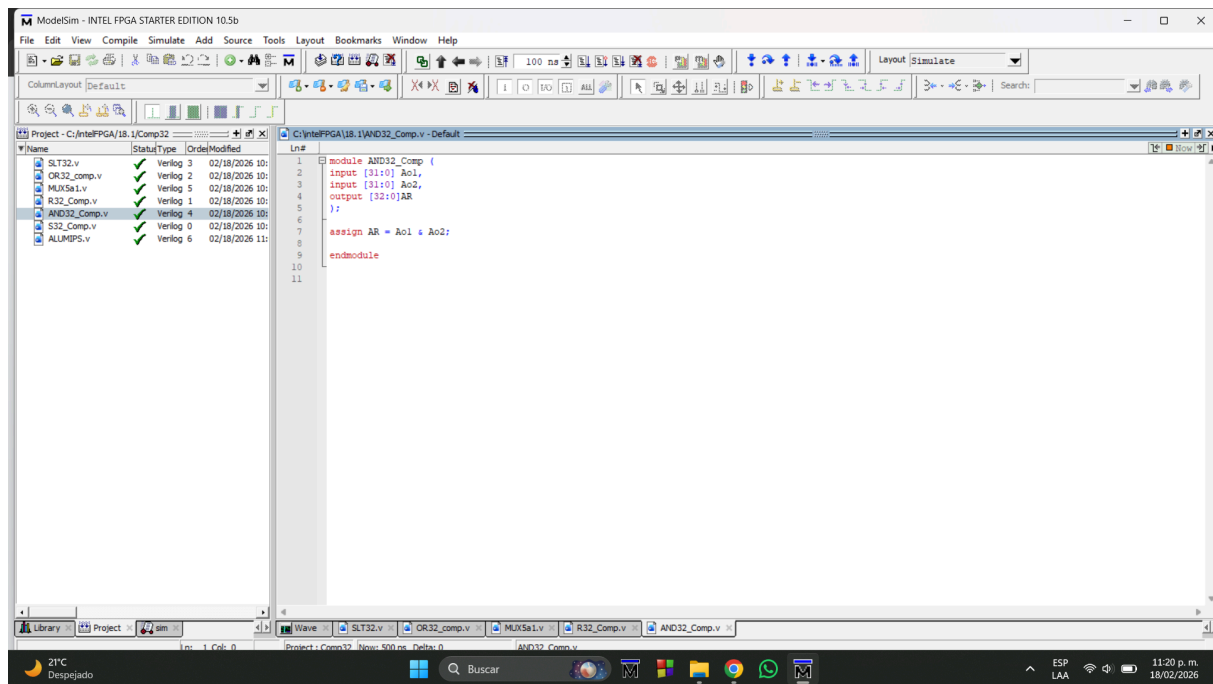


```
module R32_Comp (  
  input [31:0]Ro1,  
  input [31:0]Ro2,  
  output [32:0]RR  
);
```

```
  assign RR = Ro1 - Ro2;
```

```
endmodule
```

AND

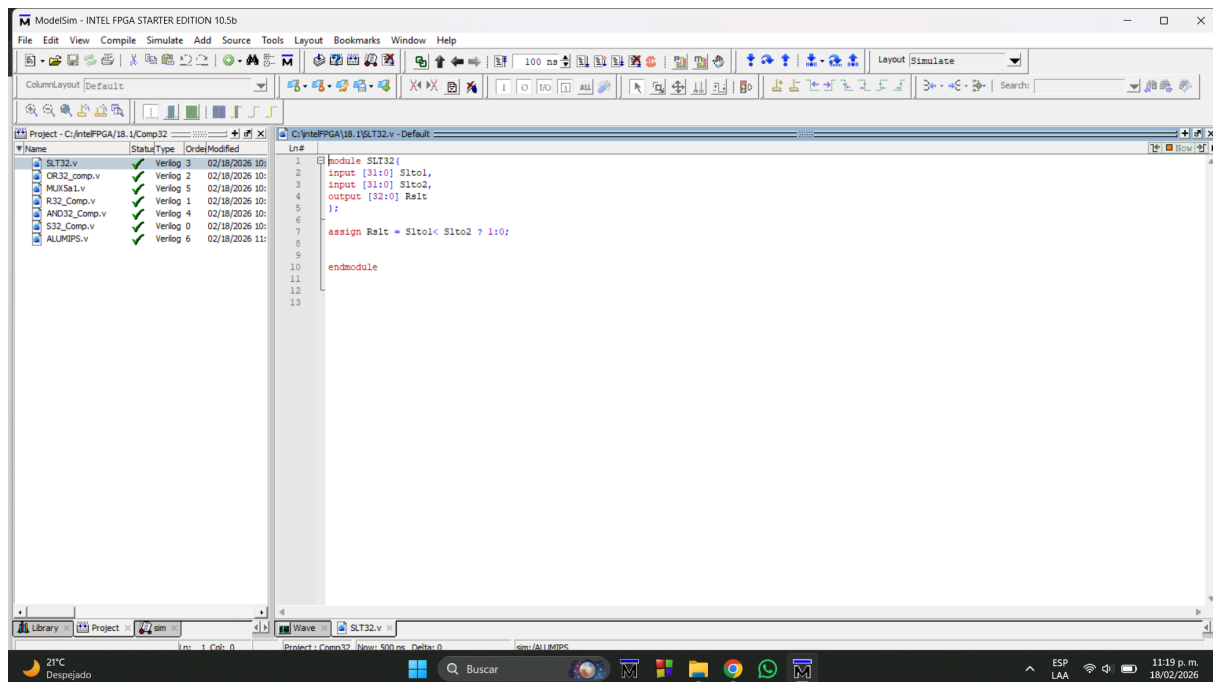


```
module AND32_Comp (  
input [31:0] Ao1,  
input [31:0] Ao2,  
output [32:0] AR  
);
```

```
assign AR = Ao1 & Ao2;
```

```
endmodule
```

SLT

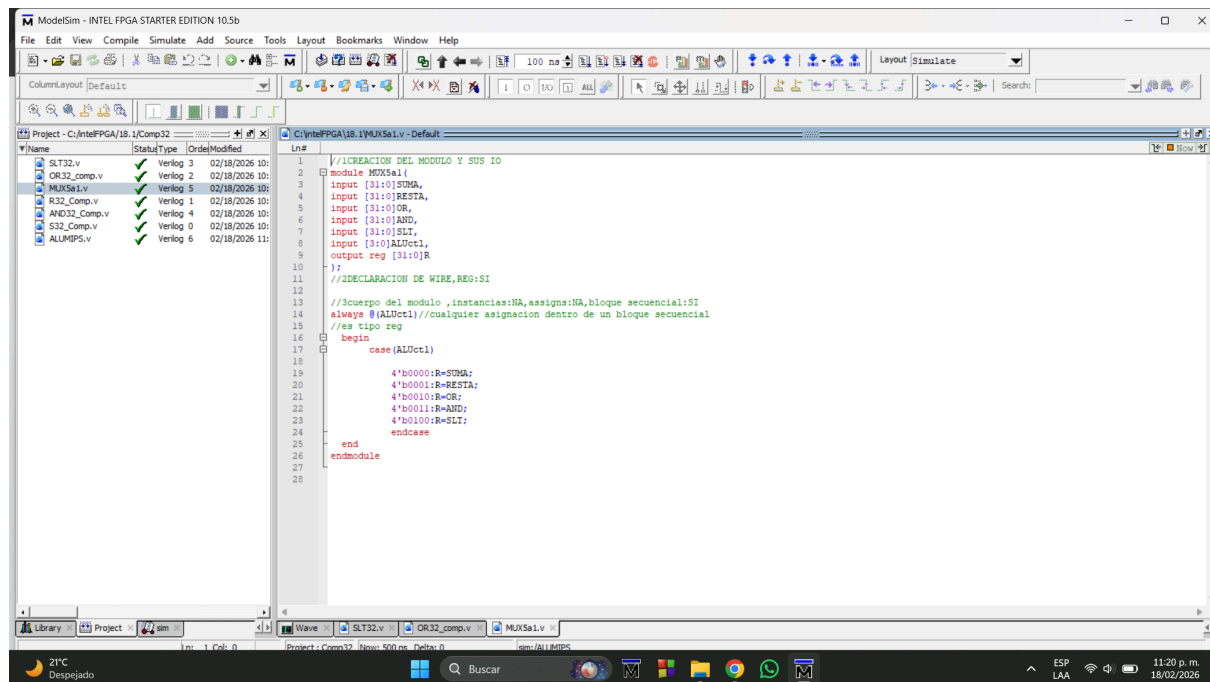


```
module SLT32(  
input [31:0] Slto1,  
input [31:0] Slto2,  
output [32:0] Rslt  
);
```

```
assign Rslt = Slto1 < Slto2 ? 1:0;
```

```
endmodule
```

MUX5a1



//1CREACION DEL MODULO Y SUS IO

```
module MUX5a1(  
input [31:0]SUMA,  
input [31:0]RESTA,  
input [31:0]OR,  
input [31:0]AND,  
input [31:0]SLT,  
input [3:0]ALUctl,  
output reg [31:0]R  
);
```

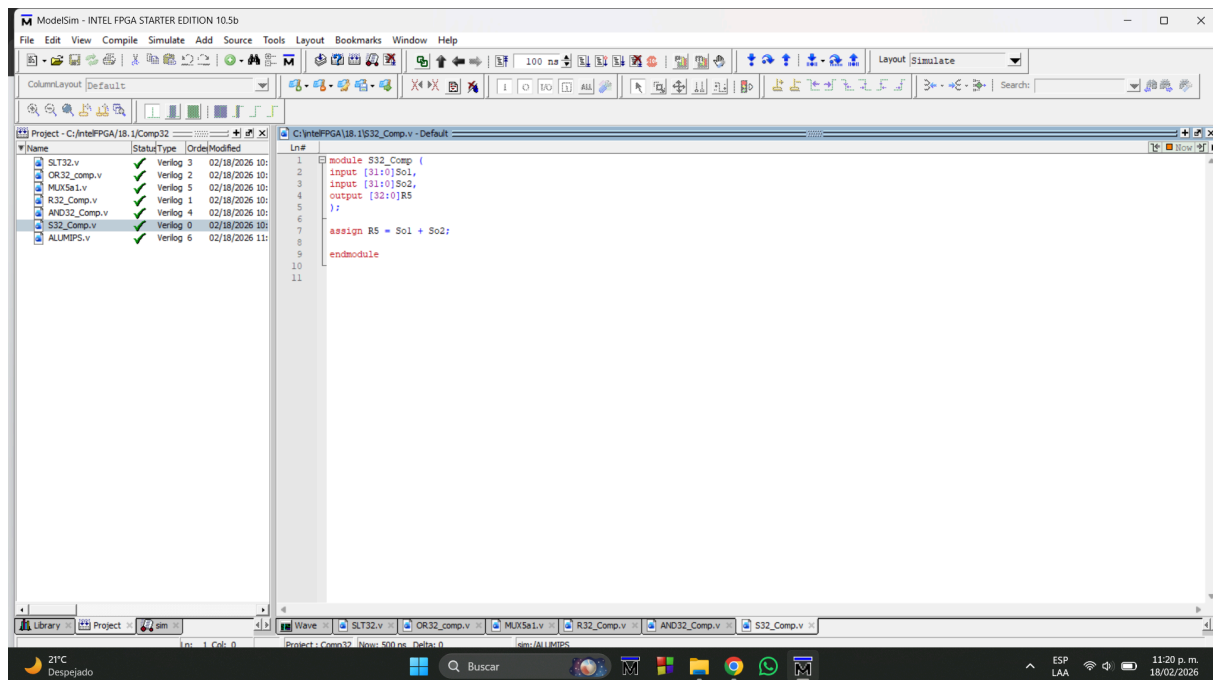
//2DECLARACION DE WIRE,REG:SI

```
//3cuerpo del modulo ,instancias:NA,assigns:NA,bloque secuencial:SI  
always @(ALUctl)//cualquier asignacion dentro de un bloque secuencial  
//es tipo reg  
begin  
    case(ALUctl)
```

```
        4'b0000:R=SUMA;  
        4'b0001:R=RESTA;  
        4'b0010:R=OR;  
        4'b0011:R=AND;  
        4'b0100:R=SLT;  
    endcase
```

```
end  
endmodule
```

Suma

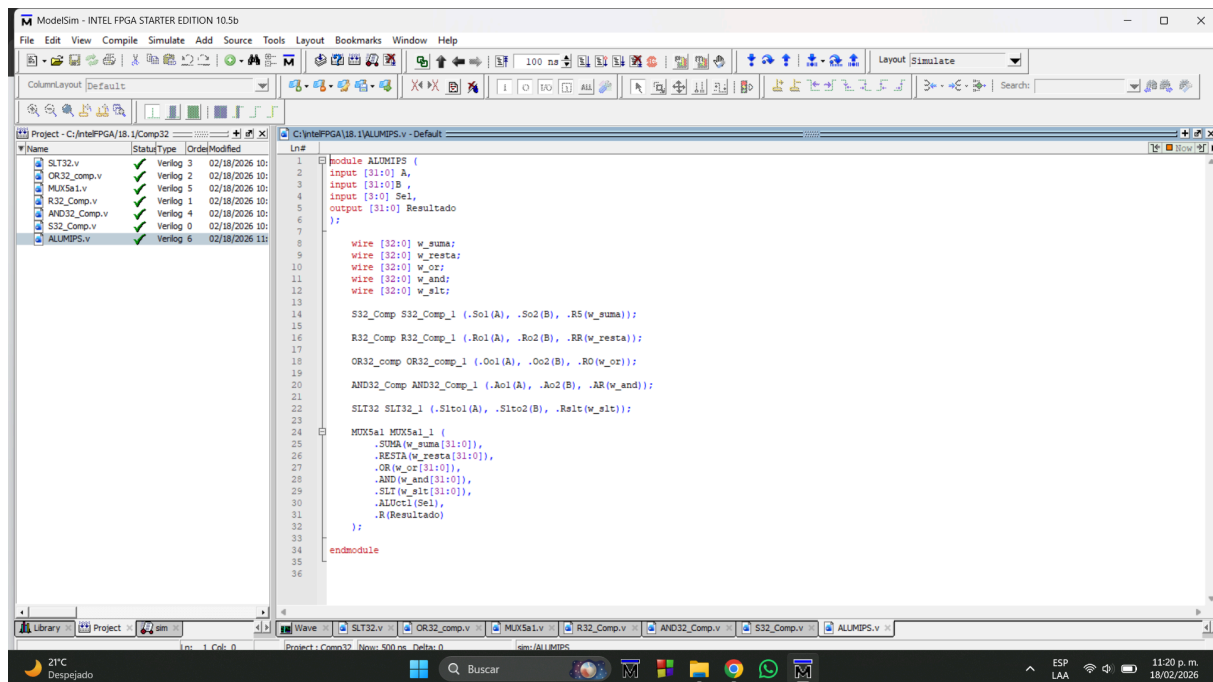


```
module S32_Comp (  
input [31:0]So1,  
input [31:0]So2,  
output [32:0]R5  
);
```

```
assign R5 = So1 + So2;
```

```
endmodule
```

ALUMIPS



```
module ALUMIPS (  
input [31:0] A,  
input [31:0] B ,  
input [3:0] Sel,  
output [31:0] Resultado  
);
```

```
wire [32:0] w_suma;  
wire [32:0] w_resta;  
wire [32:0] w_or;  
wire [32:0] w_and;  
wire [32:0] w_slc;
```

```
S32_Comp S32_Comp_1 (.So1(A), .So2(B), .R5(w_suma));
```

```
R32_Comp R32_Comp_1 (.Ro1(A), .Ro2(B), .RR(w_resta));
```

```
OR32_comp OR32_comp_1 (.Oo1(A), .Oo2(B), .RO(w_or));
```

```
AND32_Comp AND32_Comp_1 (.Ao1(A), .Ao2(B), .AR(w_and));
```

```
SLT32 SLT32_1 (.Slto1(A), .Slto2(B), .Rslt(w_slc));
```

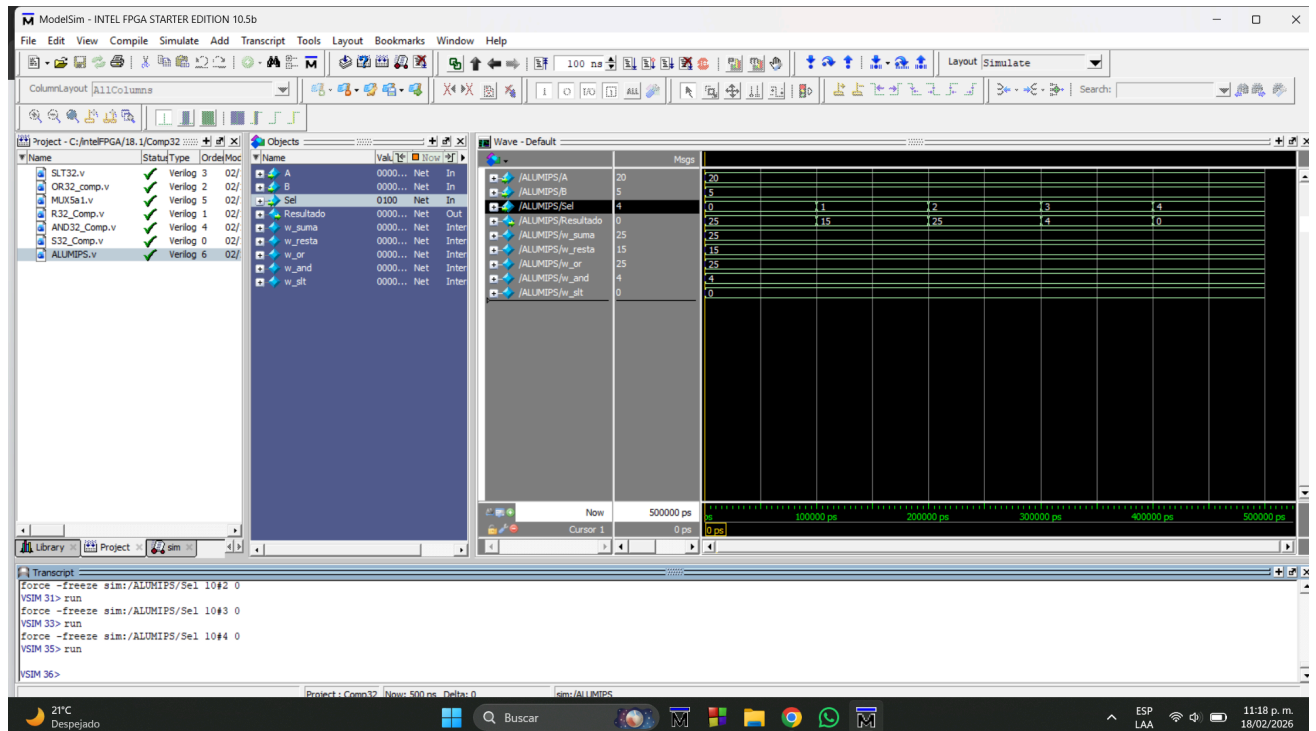
```
MUX5a1 MUX5a1_1 (  
    .SUMA(w_suma[31:0]),  
    .RESTA(w_resta[31:0]),
```



```
.OR(w_or[31:0]),  
.AND(w_and[31:0]),  
.SLT(w_sl[31:0]),  
.ALUctl(Sel),  
.R(Resultado)  
);
```

```
endmodule
```

2. Captura de resultado de suma, resta y slt con A = 20 y B = 5



Agregue, además de las solicitadas, resultados con OR y AND.